

# Template Title

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# Question 1

Given the 5 d Schwarzschild metric

$$ds^2 = - \left(1 - \frac{R_s^2}{r^2}\right) dt^2 + \left(1 - \frac{R_s^2}{r^2}\right)^{-1} dr^2 + r^2 d\Omega_3^2 \quad (1)$$

where  $R_s^2 = \frac{8GM}{3\pi}$  and  $d\Omega_3^2$  is the metric on a unit 3 sphere whose volume is  $(2\pi)^2$

## Part (a)

We have the Bekenstein–Hawking formula for the entropy of a black hole

$$S = \frac{A}{4G} \quad (2)$$

where  $A$  is the horizon area. Since we are working in 5 dimensions we need to promote the area to a volume.

$$A = R_s^3 (\text{vol}(d\Omega_3)) = \left(\sqrt{\frac{8GM}{3\pi}}\right)^3 (2\pi)^2 = 8\sqrt{\frac{2G^3 M^3 \pi}{27}} \quad (3)$$

$$S = \frac{4}{3}\sqrt{\frac{2GM^3\pi}{3}} \quad (4)$$

## Part (b)

We are tasked with computing the temperature. Note the first law of thermodynamics tells us

$$T = \frac{\mathcal{K}}{2\pi} \quad \frac{1}{T} = \frac{\partial S}{\partial M} \quad (5)$$

Thus,

$$\frac{1}{T} = \frac{\partial S}{\partial M} = 2\sqrt{\frac{2GM}{3\pi}} \quad (6)$$

$$T = \frac{1}{2}\sqrt{\frac{3}{2GM\pi}}. \quad (7)$$

We have that the surface gravity  $\mathcal{K}$  is defined by the timelike killing vector  $\xi^\mu = (\partial_t)^\mu$ . At the horizon,  $\xi^\mu$  is null, and  $\mathcal{K}$  is defined by

$$\xi^\nu \nabla_\nu \xi^\mu = \mathcal{K} \xi^\mu \quad (8)$$

We have  $\xi^\mu = (1, 0, 0, 0, 0)$ , and  $\xi_\mu = g_{\mu t} = (-f(r), 0, 0, 0, 0)$ .

$$\xi^2 = g_{\mu\nu} \xi^\mu \xi^\nu = g_{tt} = -f(r) \quad (9)$$

Now we have the contracted killing vector in terms of a known scalar we want to find a relation with  $\mathcal{K}$ .

Given the definition of  $\mathcal{K}$  (8) we can lower  $\mu$  to get  $\xi^\nu \nabla_\nu \xi_\mu = \mathcal{K} \xi_\mu$ . Then consider

$$\nabla_\mu (\xi^2) = \nabla_\mu \xi^\nu \xi_\nu = 2\xi^\nu \nabla_\mu \xi_\nu \quad (10)$$

Since  $\xi$  is a killing vector it satisfies

$$\nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu = 0 \quad \implies \quad \nabla_\mu \xi_\nu = -\nabla_\nu \xi_\mu \quad (11)$$

Thus, we have

$$\nabla_\mu (\xi^2) = -2\xi^\nu \nabla_\nu \xi_\mu \quad (12)$$

Using (8)

$$\nabla_\mu (\xi^2) = -2\mathcal{K} \xi_\mu \quad (13)$$

Consider  $\xi^2 = 0$  then its normal  $n_\mu \propto \nabla_\mu (\xi^2) \propto \xi_\mu$ .

$$\nabla_\mu (\xi^2) = -2\mathcal{K} n_\mu \quad (14)$$

Using (10), we get

$$\nabla_\mu (-\xi^2) = f'(r) \delta_\mu^r \quad (15)$$

Thus XXX

$$\mathcal{K} = \frac{1}{2} f'(r_s) \quad \text{where } f'(R_s) = \frac{2}{R_s} \quad (16)$$

$$\mathcal{K} = \frac{1}{R_s} \quad (17)$$

Thus the Hawking temperature is

$$T = \frac{\mathcal{K}}{2\pi} = \frac{1}{2\pi R_s} = \frac{1}{2\pi} \sqrt{\frac{3\pi}{8GM}} \quad (18)$$

which matches (7) hence, we have verified the first law.

## Question 2

We have the Friedmann metric

$$ds^2 = -dt^2 + a^2(t) \left[ \frac{dr^2}{1 - kr^2} + r^2 d\omega^2 \right] \quad (19)$$

Where we have the Friedmann equations

$$-\frac{\ddot{a}}{a} = \frac{4\pi G}{3} (\rho + 3\Phi) \quad (20)$$

$$\left( \frac{\dot{a}}{a} \right)^2 = \frac{8\pi G \rho}{3} - \frac{k}{a^2} \quad (21)$$

where we are given  $w = -1$  and  $\Phi = -\rho = \rho_0$  which is a constant.

We have the continuity equation

$$3 \frac{\dot{a}}{a} (\rho + \Phi) = -\dot{\rho} \quad (22)$$

which gives  $\dot{\rho} = 0$  so  $\rho = \rho_0$ .

Let us define

$$H_0^2 := \frac{8\pi G}{3} \rho_0 \quad (23)$$

which gives our Friedmann equations

$$\begin{aligned} \left( \frac{\dot{a}}{a} \right)^2 &= H_0^2 - \frac{k}{a^2} \\ \frac{\ddot{a}}{a} &= -\frac{4\pi G}{3} (\rho + 3\Phi) = H_0^2. \end{aligned}$$

Thus, we have

$$\dot{a}^2 = H_0^2 a^2 - k \quad \ddot{a} = H_0^2 a.$$

Now we can consider each case seperatly for  $k = \pm 1, 0$ .

$k = 0$

$$\left( \frac{\dot{a}}{a} \right)^2 = H_0^2 \quad \frac{\dot{a}}{a} = \pm H_0 \implies a(t) = a_* e^{H_0 t} \quad (24)$$

$k = 1$

$$\left( \frac{\dot{a}}{a} \right)^2 = H_0^2 - \frac{1}{a^2} \implies (\dot{a})^2 = H_0^2 a^2 - 1 \quad (25)$$

$$a(t) = \frac{1}{H_0} \cosh H_0 t \quad (26)$$

$$k = -1$$

$$\dot{a}^2 = H_0^2 a^2 + 1 \quad a(t) = \frac{1}{H_0} \sinh H_0 t \quad t > 0 \quad (27)$$

**See Mathematica file attached**

We see that

$$R_{\mu\nu} = -3H^2 g_{\mu\nu} \quad (28)$$

# Question 3

Given the de Sitter metric

$$ds^2 = - \left(1 - \frac{\Lambda}{3} r^2\right) dt^2 + \left(1 - \frac{\Lambda}{3} r^2\right)^{-1} dr^2 + r^2 d\Omega_2 \quad (29)$$

The time translation Killing vector is given by

$$\xi^\mu = (\partial_t)^\mu \quad \xi^2 = g_{tt} = -f(r) \quad (30)$$

where  $f(r) = 1 - \frac{\Lambda}{3} r^2$ .

$$f(r_k) = 0 \implies - \left(1 - \frac{\Lambda}{3} r_k^2\right) = 0 \quad (31)$$

Thus,

$$r_k = \sqrt{\frac{3}{\Lambda}} \quad (32)$$

$$\mathcal{K} = \frac{1}{2} \left[ \frac{2\Lambda}{3} r_k \right] = \sqrt{\frac{\Lambda}{3}} \quad T = \frac{\mathcal{K}}{2\pi} = \frac{\Lambda r_k}{6\pi} = \frac{1}{2\pi} \sqrt{\frac{\Lambda}{3}} \quad (33)$$

# Question 4

We have the FRW metric given by

$$ds^2 = -dt^2 + a^2(t) \left[ \frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right] \quad (34)$$

Closed  $k = 1$

$$ds^2 = -dt^2 + a^2(t) \left[ \frac{dr^2}{1 - r^2} + r^2 d\Omega^2 \right] \quad (35)$$

Let us define Conformal time

$$d\eta = \frac{dt}{a(t)} \quad \implies \quad dt = a(\eta) d\eta \quad (36)$$

then let  $r = \sin \chi$ , with  $0 \leq \chi \leq \pi$ . Thus  $1 - r^2 = \cos^2 \chi$  and  $dr^2 = \cos^2 \chi d\chi^2$ .

$$ds^2 = a^2(\eta) [-d\eta^2 + d\chi^2 + \sin^2 \chi d\Omega^2] \quad (37)$$

For light rays  $ds^2 = 0$  and we are told that the universe is spherical thus we can take the motion to be purely radial namely ( $d\Omega^2 = 0$ ).

Thus we have

$$d\eta^2 = d\chi^2 \quad \implies \quad \eta = \pm \chi + \text{constant} \quad (38)$$

In order to pin down the corresponding values of  $\eta$  for the Big-Bang and the Big Crunch we look at the Friedmann equations

$$\left( \frac{\dot{a}}{a} \right)^2 = \frac{8\pi G}{3} \rho - \frac{k}{a^2} \quad k = 1 \implies \left( \frac{\dot{a}}{a} \right)^2 = \frac{8\pi G}{3} \rho - \frac{1}{a^2} \quad (39)$$

Noting that for Radiation  $\rho \propto a^{-4}$ ,  $\rho = \mathcal{C}a^{-4}$  ( $\mathcal{C}$  a constant) we have

$$\begin{aligned} (\dot{a})^2 &= \frac{8\pi G \rho}{3} a^2 - 1 \\ (\dot{a})^2 &= \frac{8\pi G \mathcal{C}}{3a^2} - 1 \\ (\dot{a})^2 &= \frac{\tilde{\mathcal{C}}}{a^2} - 1. \end{aligned}$$

where  $\tilde{\mathcal{C}} := \frac{8\pi G \mathcal{C}}{3}$ .

We have

$$\begin{aligned} d\eta &= \frac{dt}{a} & dt &= a d\eta \\ \frac{da}{dt} &= \frac{\left( \frac{da}{d\eta} \right)}{\left( \frac{dt}{d\eta} \right)} = \frac{a'}{a} = \dot{a} \end{aligned}$$

.

Thus,

$$\begin{aligned}
\left(\frac{a'}{a}\right)^2 + 1 &= \frac{\tilde{C}}{a^2} \\
(a')^2 + a^2 &= \tilde{C} \\
\frac{da}{d\eta} &= \sqrt{\tilde{C} - a^2} \implies \frac{da}{\sqrt{\tilde{C} - a^2}} = d\eta \\
\eta &= \sin^{-1} \frac{a}{\sqrt{\tilde{C}}} \\
a &= \sqrt{\tilde{C}} \sin \eta.
\end{aligned}$$

Using  $dt = ad\eta$

$$\int dt = \int \sqrt{\tilde{C}} \sin \eta d\eta \quad (40)$$

$$t = -\sqrt{\tilde{C}} \cos \eta + \text{constant} \quad (41)$$

We impose that at the big bang  $t = 0$ , where  $\eta = 0$  since  $\cos 0 = 1$  so the integration constant is  $\sqrt{\tilde{C}}$  and we get

$$t = \sqrt{\tilde{C}} (1 - \cos \eta) \quad (42)$$

We see that there is a maximum at  $\eta = \frac{\pi}{2}$  and another minima at  $\eta = \pi$  corresponding to the big crunch

If we take the light ray from the big bang then the constant in (37) vanishes and we have

$$\eta = \chi \quad (43)$$

Where we have determined by the value of the big crunch is at

$$\chi = \pi \quad (44)$$

which corresponds to half the universe.